

DACT: Dynamics-Aware Learning of Coupled Torque Policies for Disturbance-Robust Legged Locomotion

Author Names Omitted for Anonymous Review. Paper-ID 2499.



Fig. 1. We present a framework for learning Dynamics-Aware Coupled Torque (**DACT**) policies for robust legged locomotion under diverse external disturbances. **DACT** learns dynamics-aware coupled torque policies that combine gait-regulating position control with feedforward torque compensation, enabling robust legged locomotion under terrain variation, payload disturbances, and external torso wrenches.

Abstract—Robust quadruped deployment requires policies that reject disturbances from payloads, terrain variation, and torso-level external forces and moments. However, standard position-control policies compensate only indirectly through fixed PD feedback, while direct-torque policies often lose high-bandwidth stabilization, and auxiliary compensation methods add additional training/compute without resolving the underlying action space-dynamics representation mismatch. We introduce DACT, a dynamics-aware framework that embeds disturbance compensation directly into the locomotion policy via coupled feedforward-torque and feedback-stabilized position actions. DACT learns a privileged-dynamics-aware latent context through next-step privileged observation reconstruction and uses physics-informed neural network (PINN) regularization to bias actions toward physically consistent whole-body stabilization. We evaluate DACT across payload variation, CoM shifts, complex terrain, and 6-DOF torso wrenches, with real-world tests including dynamic liquid payload transport across unstructured outdoor terrain. Results show that DACT improves disturbance rejection and locomotion stability over position-control, direct-torque, and adaptive torque-compensation baselines, enabling robust real-world locomotion under diverse external disturbances.

I. INTRODUCTION

Quadruped robots have achieved substantial advances in locomotion capabilities [1]–[5], enabling agile mobility through unstructured environments that remain difficult or inaccessible for wheeled platforms. Reinforcement learning (RL) has further expanded these capabilities by training robust

gait policies in large-scale simulation, reducing reliance on hand-designed models while supporting dynamic behaviors across diverse terrains and commands. However, real-world deployments rarely occur under nominal conditions: robots may carry supplies over uneven terrain, recover from contact with obstacles or humans, or maintain balance as onboard loads shift during motion. These disturbances perturb body motion, reshape contact forces, and disrupt the coupling between foot placement, force generation, and torso stabilization. Robust deployment therefore requires policies that reject disturbances online while preserving stable and reliable locomotion across diverse operating conditions.

Payloads and external disturbances alter the robot’s effective locomotion dynamics through the introduction of unmodeled forces and moments. Most RL policies handle these effects indirectly with joint-position targets tracked by low-level PD controllers [3], [4], [6]–[9]; while effective, this fixed impedance forces disturbance rejection to emerge through changes in desired joint kinematics rather than direct compensatory torques. This creates a representational mismatch between the action space and the force generation governing balance and whole-body stabilization. Direct-torque policies reduce this mismatch [10]–[12], but sacrifice the stabilizing structure of feedback control. Recent auxiliary compensation methods add learned or control-theoretic adaptive joint-position/torque offsets to frozen nominal policies [8],

[13]–[15]. However, these methods treat compensation as an external post-hoc module, requiring additional computation or training while leaving the base policy unable to encode disturbance-dependent dynamics or produce actions intrinsically aligned with disturbed whole-body stabilization. In contrast, biological and model-based controllers couple feedforward force generation with feedback stabilization for load- and contact-dependent adaptation [16]–[22], motivating coupled torque–position control policies.

To address these fundamental limitations, we introduce *Dynamics-Aware Coupled Torque (DACT)* policies, a dynamics-aware learning framework that embeds disturbance compensation directly within the locomotion policy. **DACT** couples feedforward joint torques with feedback-stabilized joint-position commands, enabling direct compensatory force generation while preserving high-frequency posture stabilization. To handle the partial observability of external disturbances, **DACT** trains a latent context encoder to reconstruct next-step privileged observations, encouraging the encoder to capture disturbance-dependent changes in the robot’s effective dynamics. Additionally, a physics-informed neural network (PINN) regularization loss further biases learned actions toward physically consistent whole-body stabilization. Together, these components bridge gait-level policy learning and force-level disturbance rejection, producing a unified policy for robust locomotion under payload, terrain, and torso-level perturbations. The main contributions of this paper are:

- We introduce *Dynamics-Aware Coupled Torque (DACT)* policies, a novel dynamics-aware legged locomotion policy learning framework that internalizes disturbance compensation through coupled torque–position actions, privileged dynamics reconstruction, and PINN-based whole-body action regularization.
- We show that these components provide complementary benefits for disturbance rejection: in simulation, **DACT** reduces failures by 43.0% under payload/CoM shifts and 49.6% under 6-DOF torso wrenches relative to the next-best method, while achieving the strongest overall torso stability.
- We demonstrate sim-to-real transfer on a Unitree Go1, showing robust locomotion across real-world terrains and disturbances, including dynamic liquid payload transport as a challenging external perturbation.

II. RELATED WORK

Disturbance-Robust Legged Locomotion. Robust legged locomotion has been addressed with model-based, learning-based, and hybrid methods. Model-based approaches combine MPC/QP control with inertial identification [23], [24], CLFs [19], L1 or reduced-order adaptive control [21], [25], and chance-constrained disturbance models [26], but often require substantial computation, modeling, and tuning. Learning-based methods use domain randomization and privileged RL to infer latent physical parameters [4], [6], [27], learn axillary residual corrections for frozen policies [8], [13], or explicitly estimate disturbances such as torso forces or payload states [9].

Hybrid methods combine RL with MPC residuals, MPC-generated torques, or adaptive torque compensation [15], [22], [28]–[30], but introduce additional modules, training, and runtime overhead. In contrast, **DACT** internalizes disturbance compensation within a single locomotion policy through dynamics-aware training objectives.

Action Spaces for Legged Locomotion. Model-based controllers often pair feedforward torques for stance force regulation with PD feedback for swing tracking and posture stabilization [1], [2]. Most RL policies instead output joint-position targets tracked by fixed PD controllers [3], [4], [6]–[9], which is effective but models and controls interaction forces only indirectly through joint-target errors. Direct-torque policies provide more immediate force control [10]–[12], but are harder to train and lack high-bandwidth feedback stabilization. Motivated by model-based control, **DACT** learns coupled torque–position actions for both feedforward compensation and feedback stabilization.

Physics-Informed Neural Networks (PINN). PINNs enforce dynamics consistency by embedding governing equations into learning objectives [31], [32]. They are widely used in fluid dynamics, heat transfer, and solid mechanics [33], [34], and increasingly in robotics for manipulator dynamics, quadrotor aerodynamics, soft robots, state estimation, and motion planning [32], [35]–[40]. However, PINNs remain underexplored for disturbance-aware quadruped policy learning, where whole-body dynamics consistency can provide a principled inductive bias.

III. APPROACH

A. Preliminaries

a) *Quadruped Robot Dynamics:* Let $\rho = [\mathbf{p}^\top, \boldsymbol{\Theta}^\top, \mathbf{q}^\top]^\top$ denote the floating-base torso position, orientation, and joint configuration, with n actuated joints. Define the actuation-selection matrix $\mathbf{S} = [\mathbf{0}_{n \times 6} \ \mathbf{I}_n]$, so that $\mathbf{S}^\top \boldsymbol{\tau}^{cmd}$ maps commanded joint torques into generalized forces with zero actuation at the six floating-base coordinates. The full-body dynamics are [1], [2]:

$$\mathbf{S}^\top \boldsymbol{\tau}^{cmd} + \mathbf{J}_c^\top \mathbf{f}^r = \mathbf{A}\ddot{\rho} + \mathbf{c}\dot{\rho} + \mathbf{g}, \quad (1)$$

where $\mathbf{A}$, $\mathbf{c}$, and $\mathbf{g}$ are the inertia matrix, Coriolis matrix, and gravity vector, respectively, and $\mathbf{J}_c^\top \mathbf{f}^r$ maps ground reaction forces (GRFs) to generalized forces. Most RL locomotion policies [7] output desired joint positions $\mathbf{a}$ and rely on a low-level PD controller to generate joint torques, $\boldsymbol{\tau}^{cmd} \approx \mathbf{K}_p(\mathbf{a} - \mathbf{q}) - \mathbf{K}_d\dot{\mathbf{q}}$, where $\mathbf{K}_p$ and $\mathbf{K}_d$ are proportional and derivative gain matrices. External loading and disturbance-dependent contact changes contribute generalized forces $\Delta\boldsymbol{\tau} = \mathbf{J}_c^\top \Delta\mathbf{f}^r$, where $\Delta\mathbf{f}^r$ denotes GRF variations relative to the nominal contact forces [15]. The disturbed PD-controlled dynamics are:

$$\mathbf{A}\ddot{\rho} + \mathbf{c}\dot{\rho} + \mathbf{g} - \mathbf{J}_c^\top \mathbf{f}^r - \Delta\boldsymbol{\tau} \approx \mathbf{S}^\top [\mathbf{K}_p(\mathbf{a} - \mathbf{q}) - \mathbf{K}_d\dot{\mathbf{q}}] \quad (2)$$

B. Problem Formulation

A quadruped observes proprioceptive measurements from joint encoders and an IMU, including joint states $(\mathbf{q}, \dot{\mathbf{q}})$,

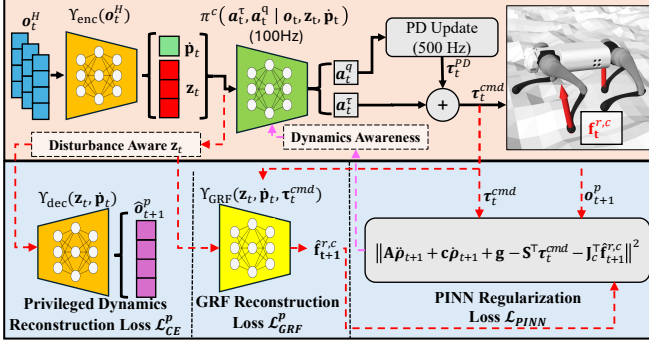


Fig. 2. Overview of **DACT**. A privileged-dynamics-aware encoder maps observation history to a latent context for a coupled torque–position policy and PINN regularization trains the policy to produce physically consistent feedforward compensation and feedback-stabilized locomotion under diverse disturbances.

torso orientation (Θ), and angular velocity ($\dot{\Theta}$). Given commanded torso velocities $\mathbf{c} = [\dot{p}_x^{cmd}, \dot{p}_y^{cmd}, \dot{\Theta}_{yaw}^{cmd}]^\top$, the policies observation is $\mathbf{o}_t = [\mathbf{c}_t, \mathbf{g}_t^p, \dot{\Theta}_t, \mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}]^\top$, where $\mathbf{g}_t^p$ is the gravity vector projected by Θ_t and $\mathbf{a}_{t-1}$ is the previous position-control action. A context encoder Υ_{enc} maps an H -step observation history to a latent context and base-velocity estimate, $\Upsilon_{enc}(\mathbf{o}_t^H) \mapsto (\mathbf{z}_t, \hat{\mathbf{p}}_t)$, following [27]. A decoder Υ_{dec} reconstructs the next observation, $\Upsilon_{dec}(\mathbf{z}_t, \hat{\mathbf{p}}_t) \mapsto \mathbf{o}_{t+1}$, encouraging $\mathbf{z}_t$ to encode temporal dynamics. During training, an asymmetric critic receives a privileged observation

$$\mathbf{o}_t^p = [\mathbf{o}_t, p_z, \hat{\mathbf{p}}_t, \mathbf{I}_{ext}, \Delta m_{robot}, \Delta \text{CoM}, \Delta K_{p,d}, \mathbf{h}]^\top \quad (3)$$

which augments $\mathbf{o}_t$ with the measured torso height and velocity, external torso disturbance wrench impulses, feet height and contact state, domain randomization values, and robot centric height-map.

Under this formulation, we learn a disturbance-adaptive policy $\pi(\mathbf{a}^q | \mathbf{o}_t, \mathbf{z}_t, \hat{\mathbf{p}}_t)$ that outputs joint-position actions added to the nominal stance $\mathbf{q}^{\text{stand}}$ to form PD targets $\mathbf{q}_t^{\text{ex}}$ in Eq. (2). Encoding persistent disturbance compensation through these targets requires a gain-dependent offset $\Delta \mathbf{q}_{des} = \mathbf{K}_p^{-1} \Delta \tau$ under ideal PD control. However, encoding large, persistent compensatory torques results in exaggerated position targets displaced from the intended gait kinematics, requiring the same action representation to encode both desired motion and tracking errors introduced solely to generate force. This gain-dependent entanglement can complicate learning the relationship between intended motion and required force, motivating our coupled action space, which represents disturbance-compensating torques explicitly alongside gait-regulating kinematic targets.

C. Disturbance-Aware Coupled Torque Policy

To separate intended motion from disturbance compensation, we introduce a coupled policy $\pi^c(\mathbf{a}_t^q, \mathbf{a}_t^\tau | \mathbf{o}_t, \mathbf{z}_t, \hat{\mathbf{p}}_t)$ that jointly outputs feedforward joint torques $\mathbf{a}_t^q$ and position actions $\mathbf{a}_t^\tau$. The resulting commanded torque and coupled robot-disturbance dynamics is:

$$\mathbf{A}\ddot{\mathbf{p}} + \mathbf{c}\dot{\mathbf{p}} + \mathbf{g} - \mathbf{J}_c^\top \mathbf{f}^r - \Delta \tau \approx \mathbf{S}^\top [\mathbf{a}^\tau + \mathbf{K}_p(\mathbf{a}^q - \mathbf{q}) - \mathbf{K}_d \dot{\mathbf{q}}] \quad (4)$$

The torque channel provides explicit joint-torque compensation for disturbances $\Delta \tau$ in Eq. (2), while position targets generate PD torques for gait regulation and high-frequency feedback stabilization. This decomposition allows persistent, low-frequency compensation through feedforward torques while retaining rapid feedback correction of tracking and posture errors, without requiring compensation to be encoded as displaced kinematic targets. By separating these roles in the action representation, the policy can learn force compensation independently of the gain-dependent target offsets required by position-only control. It therefore serves as a structural inductive bias aligning policy learning with force generation and gait regulation.

However, the disturbance-induced contact dynamics $\mathbf{J}_c^\top \Delta \mathbf{f}^r$ remain partially observable to π^c throughout training because proprioception does not directly measure $\Delta \mathbf{f}^r$; making the coupled robot–disturbance–terrain contact dynamics weakly identifiable from $\mathbf{o}_t$ or $\mathbf{o}_t^p$ alone. We therefore augment the asymmetric critic’s privileged input with measured GRFs, $\mathbf{f}^{r,c} = \mathbf{f}^r + \Delta \mathbf{f}^r$, providing direct contact information for value estimation during training.

To further encode these dynamics in the policy context, we train the history encoder Υ_{enc} to produce a latent representation $\mathbf{z}_t$ that reconstructs the next privileged observation $\mathbf{o}_{t+1}^p$, excluding GRFs, while a separate GRF decoder Υ_{GRF} predicts next-step contact forces conditioned on both encoder outputs and the commanded torque τ_t^{cmd} resulting from the coupled action output:

$$\begin{aligned} \Upsilon_{enc}(\mathbf{o}_t^H) &\mapsto (\mathbf{z}_t, \hat{\mathbf{p}}_t), \Upsilon_{dec}(\mathbf{z}_t, \hat{\mathbf{p}}_t) \mapsto \hat{\mathbf{o}}_{t+1}^p, \\ \Upsilon_{GRF}(\mathbf{z}_t, \hat{\mathbf{p}}_t, \tau_t^{cmd}) &\mapsto \hat{\mathbf{f}}_{t+1}^{r,c} \end{aligned} \quad (5)$$

Here, τ_t^{nom} combines the feedforward and nominal PD torques of the causally applied action. Its recorded value is normalized and treated as constant during reconstruction, allowing supervision to update the encoder and decoders without updating the actor through the torque input. Targets are aligned to the corresponding transition and exclude invalid reset crossings. The privileged reconstruction objective is

$$\mathcal{L}_{PCE} \|\hat{\mathbf{o}}_{t+1}^p - \mathbf{o}_{t+1}^p\|_2^2 + \mathcal{L}_{GRF} + \|\hat{\mathbf{p}}_t - \hat{\mathbf{p}}_t\|_2^2 + \mathcal{L}_{KD}. \quad (6)$$

where reconstruction errors use normalized quantities, $\mathcal{L}_{GRF}$ supervises predicted forces against measured GRFs, and $\mathcal{L}_{KD}$ is the VAE Kullback–Leibler regularizer [27]. This objective encourages the context to represent disturbance and terrain dynamics while explicitly relating coupled actions to contact-force predictions. However, privileged supervision alone does not ensure physically meaningful use of the coupled dynamics; without an additional dynamical bias, policy optimization may still exploit artifacts that are predictive in simulation but inconsistent with the whole-body mechanics.

D. Dynamics-Aware Learning via PINN Regularization

To align π^c with the dynamics of the disturbed robot system, we introduce a PINN auxiliary loss that regularizes policy learning through whole-body dynamics consistency. Beyond task rewards, this loss biases PPO updates toward coupled torque–position commands and predicted contact forces

TABLE I

OF FAILURES, COMMAND TRACKING, AND TORSO STABILITY PERFORMANCE ACROSS VARIOUS TERRAINS AND DISTURBANCE CONDITIONS. 1,200 SUCCESSFUL EPISODES ARE COLLECTED PER-DISTURBANCE CONDITION. METRICS REPORTING NORMS USE THE L1 NORM AND ALL OTHER METRICS REPORT MAE. BEST RESULTS ARE BOLDED AND SECOND-BEST RESULTS ARE UNDERLINED.

Metric	No Disturbance							Payload and CoM Shifts							External 6DOF Wrenches						
	DW	TO	A1	A2	A3	DC	RA	DW	TO	A1	A2	A3	DC	RA	DW	TO	A1	A2	A3	DC	RA
FR↓	1.00%	0.66%	0.58%	0.66%	<u>0.41%</u>	0.33%	0.57%	32.8%	37.8%	25.6%	36.7%	<u>23.7%</u>	15.1%	34.8%	18.7%	30.9%	<u>17.6%</u>	19.7%	19.1%	9.71%	43.2%
$\mathbf{v}_{x,y}^e \downarrow$	0.124	0.130	<u>0.126</u>	<u>0.126</u>	0.130	0.127	0.154	0.179	0.189	0.170	<u>0.169</u>	0.171	0.165	0.209	0.314	0.332	0.317	0.312	<u>0.309</u>	0.306	0.344
$\omega_{yaw}^e \downarrow$	0.157	0.127	<u>0.138</u>	0.138	0.150	<u>0.136</u>	0.163	0.229	<u>0.185</u>	0.196	0.191	0.194	0.176	0.253	0.278	<u>0.227</u>	0.238	0.237	0.244	0.222	0.306
$\mathbf{q}_j^e \downarrow$	0.169	—	0.091	0.101	0.106	<u>0.097</u>	0.149	0.204	—	<u>0.114</u>	0.119	0.124	0.113	0.167	0.192	—	0.108	0.113	0.117	<u>0.109</u>	0.176
$p_x^e \downarrow$	0.161	0.158	0.158	<u>0.156</u>	0.155	0.158	0.155	0.170	0.182	0.167	<u>0.166</u>	0.168	0.164	0.169	0.173	0.170	<u>0.168</u>	<u>0.168</u>	<u>0.168</u>	0.165	0.175
$\ \mathbf{g}\ _1 \downarrow$	0.088	0.087	0.099	0.099	0.116	0.092	0.097	0.176	0.175	0.150	<u>0.145</u>	0.152	0.122	0.198	0.141	0.139	<u>0.134</u>	<u>0.134</u>	0.148	0.120	0.161
$v_z \downarrow$	0.073	0.077	0.075	0.067	0.070	<u>0.068</u>	0.084	0.108	0.111	0.098	0.094	<u>0.093</u>	0.085	0.132	0.148	0.153	0.130	<u>0.120</u>	0.124	0.113	0.172
$\ \omega\ _1 \downarrow$	0.651	0.609	0.631	0.629	0.621	0.593	0.641	0.873	0.843	0.822	0.805	<u>0.761</u>	0.688	1.082	1.123	0.987	1.024	0.998	<u>0.966</u>	0.939	1.211
$\ \tau \mathbf{q}\ _1 \downarrow$	43.61	46.24	41.82	43.36	39.36	<u>41.32</u>	44.26	50.97	56.88	50.42	49.74	45.33	<u>48.29</u>	60.14	71.36	79.88	69.68	67.42	61.89	<u>64.69</u>	73.26
P.A.↑	—	—	-0.68	0.777	0.804	0.761	0.122	—	—	-0.471	0.767	0.794	0.754	0.288	—	—	-0.531	0.771	0.794	0.754	0.204

that provide a physically plausible explanation of observed motion under external disturbances. Given the actor’s current commanded torque τ_t^{cmd} , the observed transition summarized by $\mathbf{o}_{t+1}^p$, and the predicted contact forces $\hat{\mathbf{f}}_{t+1}^{r,c}$ conditioned on the learned context and commanded torque, we construct the loss from the following dynamics residual:

$$\mathcal{L}_{PINN} = \left\| \mathbf{A}_r \hat{\mathbf{p}}_{t+1} + \mathbf{c}_r \dot{\mathbf{p}}_{t+1} + \mathbf{g}_r - \mathbf{S}^\top \tau_t^{cmd} - \mathbf{J}_c^\top \hat{\mathbf{f}}_{t+1}^{r,c} \right\|^2, \quad (7)$$

where $\hat{\mathbf{p}}_{t+1}$ is estimated by backward finite differences of simulator-reported velocities over the completed control interval.

Observed accelerations and rigid-body dynamics quantities remain fixed during optimization, while gradients propagate through the actor’s commanded torque and the torque-conditioned GRF predictions. This couples the learned context and both action channels to their predicted contact response, encouraging PPO updates that remain compatible with observed motion. The GRF pathway additionally makes the six unactuated base equations informative for learning, allowing the full 18-dimensional residual to shape representations of whole-body contact dynamics rather than only actuated joint behavior. The GRF decoder remains trainable through measured-force reconstruction, but its parameters are held fixed during physics updates, directing dynamics regularization toward the encoder and actor without directly modifying decoder weights to absorb force-balance errors. Together, these mechanisms provide a physics-guided inductive bias linking inferred context, torque generation, and contact-mediated motion, supporting coordinated disturbance compensation and posture regulation. Implementation and gradient-handling details are provided in the appendix.

E. Training DACT

Fig. 2 summarizes DACT. We train π^c with PPO using $\mathcal{L} = \mathcal{L}_{PPO} + \mathcal{L}_{PCE} + \mathcal{L}_{PINN}$, where $\mathcal{L}_{PPO} + \mathcal{L}_{PINN}$ updates both π^c and $\mathbf{Y}_{enc}$, while $\mathcal{L}_{PCE} + \mathcal{L}_{PINN}$ updates $\mathbf{Y}_{enc}$, $\mathbf{Y}_{dec}$, and $\mathbf{Y}_{grf}$ in a separate step.

a) *Torque coupling rewards.*: To encourage coherent use of the coupled action space, we regularize the interaction between feedforward torques $\mathbf{a}^\tau$ and feedback PD torques τ^{fb} . Let $\cos_sim(\cdot, \cdot)$ denote cosine similarity. We reward

TABLE II

TRAINING AND TESTING RANGES FOR PAYLOAD AND DISTURBANCE RANDOMIZATION. CoM VALUES ARE SYMMETRIC DISPLACEMENT LIMITS; WRENCH VALUES ARE PLANAR, VERTICAL, AND ANGULAR VELOCITY PERTURBATION LIMITS.

Parameters	Training Range	Testing Range
Payload (kg)	[−1.0, 8.0]	[−3.0, 12.0]
Center of Mass (m)	[0.20, 0.15, 0.15]	[0.30, 0.225, 0.225]
Max. Wrench Velo.	[1.50, −0.50, 1.50]	[2.25, −0.75, 2.25]

cooperative torque alignment and penalize opposing torques via

$$r_{align} = \max\left(0, \cos_sim(\mathbf{a}^\tau, \tau^{fb})\right), \quad (8)$$

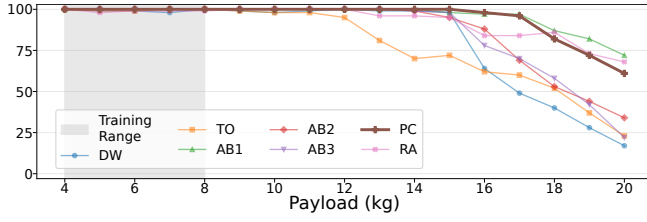
$$r_{conflict} = \frac{m_r}{m_r + \Delta m} \min\left(0, \cos_sim(\mathbf{a}^\tau, \tau^{fb})\right).$$

where m_r is robot mass and Δm is added payload mass. The mass scaling relaxes the conflict penalty for heavier payloads, allowing stronger corrective torques when larger disturbance compensation is required.

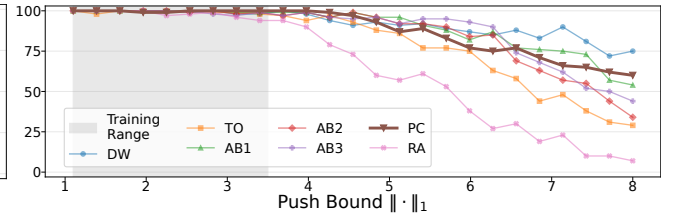
b) *Curriculum Learning*: We train DACT with a two-stage curriculum. In **Stage I**, *Dynamics-Aware Policy Discovery*, π^c is trained with standard locomotion rewards [27], narrow randomization of torso mass Δm and CoM offset ΔCoM , and small impulsive 6DOF torso wrenches.

In **Stage II**, *Disturbance Adaptation*, we gradually widen the range of domain randomization using a velocity-tracking performance gated linear-update rule. The Stage I reward structure is retained, with stronger penalties on torso orientation error, height deviation, and undesired angular velocity to promote stabilization under larger disturbances.

Both stages use $\mathcal{L}_{PINN}$ and $\mathcal{L}_{PCE}$. We compute $\mathbf{A}$, $\mathbf{c}$, $\mathbf{g}$, and $\mathbf{J}_c$ for the PINN loss with Pinocchio [41], applying each sampled Δm and ΔCoM to the robot model before dynamics evaluation so the residual reflects the randomized dynamics. Impulse wrenches are captured through the observed 6DOF torso velocity. Full reward, domain-randomization, and PINN-loss details are provided in the supplementary material.



(a) Increasing Centered Payload Mass Success Rates.



(b) Increasing External Torso Wrench Success Rates.

Fig. 3. Episodic success rates across under increasing payload mass and external wrench magnitude.

IV. EXPERIMENTS

A. Experiment Setup

Comparison Methods and Implementation Details:

We evaluate **DACT** in simulation and on a Unitree Go1. Policies are trained in Genesis [42] using an open-source implementation of [7], for 8,000 iterations: 2,000 in Stage I and 6,000 in Stage II. We compare **DACT** (**DC**) against position control **DW** [27], feedforward torque control **TO** [10], and adaptive-control-based torque compensation **RA** [15] methods. We also evaluate three ablations: **A1** removes coupled-action rewards, privileged dynamics information, and PINN regularization; **A2** adds only coupled-action rewards; and **A3** adds privileged critic inputs and reconstruction but omits PINN regularization. All methods use the same backbone MLP architecture, rewards, curriculum, random seed, 100 Hz policy rate, and 500 Hz PD loop with $K_p = 30.0$ and $K_d = 0.65$. Full implementation details are provided in the supplementary material.

Evaluation Metrics: We evaluate simulated performance using four metric groups. **Success** reports the episodic failure rate (FR), where failure occurs when absolute roll/pitch exceeds 1.57 rad or the torso contacts the ground. **Command tracking** reports MAE planar $\mathbf{v}_{x,y}^e$, and yaw ω_{yaw}^e velocity tracking, and joint-position tracking $\mathbf{q}_j^e$. **Torso stability** reports MAE in torso height p_z^e , the mean horizontal gravity projection $\|\mathbf{g}\|_1$, MAE vertical velocity v_z , and mean roll/pitch angular velocity $\|\omega_{x,y}\|_1$. **Energy** reports mean joint mechanical power $\|\tau^{cmd}\dot{\mathbf{q}}\|_1$. For **DACT** and its ablations, we additionally measure coordination between the coupled outputs via the Power Alignment (**P.A.**) metric: $\cos_{sim}(\psi^q, \psi^\tau)$, where $\psi^q = \tau^{PD}\dot{\mathbf{q}}$ and $\psi^\tau = \mathbf{a}^\tau\dot{\mathbf{q}}$.

B. Simulated Results

Disturbance Rejection in Simulation: We evaluate all methods across rough heightfields (± 0.08 m), 20° slopes, stairs (0.40 m tread, 0.10 m height), discrete obstacles up to 0.10 m, and wave terrain (0.20 m amplitude, two cycles). Each method runs 200 episodes per terrain with velocity commands sampled from $[-1.0, 1.0]$ every 10 s. Table I reports performance averaged across terrains under no disturbance, payload/CoM shifts, and external 6-DOF torso wrenches; disturbance ranges are given in Table II. **DACT** achieves the strongest off-nominal robustness while preserving nominal locomotion, where all methods perform similarly. Under payload/CoM shifts and 6-DOF wrenches, **DACT** reduces failures by 43.0% and 49.6% relative to the next-

TABLE III

RECONSTRUCTION AND DYNAMICS-CONSISTENCY RMSE ON ROUGH TERRAIN, WITH 100 EPISODES COLLECTED PER ROLLOUT SOURCE AND CONDITION. BOLD INDICATES LOWER ERROR WITHIN EACH SOURCE AND CONDITION.

Metric	Payload + CoM				Impulse + CoM			
	A3 rollouts		DC rollouts		A3 rollouts		DC rollouts	
	A3	DC	A3	DC	A3	DC	A3	DC
Δm (kg)	4.073	1.594	4.354	1.470	3.158	4.696	3.031	1.934
ΔCoM (m)	0.1009	0.0580	0.1122	0.0456	0.1158	0.0760	0.1301	0.0625
$\hat{\mathbf{o}}_{t+1}$	0.1766	0.1487	0.2231	0.0991	0.1523	0.1421	0.2528	0.0961
Dynamics residual	0.6469	0.6298	0.6661	0.5394	0.5529	0.5608	0.6698	0.4331
GRF (N)	21.788	19.473	24.277	12.898	17.079	17.357	21.062	12.683
Linear, 50 ms	1.0290	0.9521	1.2294	0.9226	0.8167	0.9070	1.1883	0.8047
Linear, 100 ms	1.6293	1.5962	2.1820	1.5202	1.3440	1.5841	2.0203	1.3160
Angular, 50 ms	0.3034	0.2917	0.3895	0.2665	0.2473	0.2597	0.2913	0.2170
Angular, 100 ms	0.3403	0.5476	0.4820	0.4412	0.3556	0.3833	0.5176	0.2920

best method, respectively, while achieving the best disturbed-setting torso stability and command tracking.

The baselines highlight the limitations of each action structure. **DW** is competitive nominally, but its position-only action space limits adaptation to strong perturbations. **TO** commands torques directly, but its lack of high-bandwidth feedback stabilization, leads to more failures and torso motion under severe disturbances. **RA** adds adaptive torque corrections to a frozen position policy, but terrain-induced tracking errors can trigger abrupt over-correction that destabilizes the nominal gait. In contrast, **DACT** learns compensation and stabilization jointly within a single policy. The ablations show that each **DACT** component contributes to robustness. Coupled-action rewards shift power alignment from negative in **A1** to strongly positive in **A2**, **A3**, and **DACT**, indicating coordinated rather than opposing PD and feedforward power injection. Performance improves from **A1** to **A3**, while full **DACT** achieves the lowest failures and strongest stability, showing that coupled rewards, privileged dynamics reconstruction, and PINN regularization provide complementary benefits for disturbance rejection across complex terrain.

Success Rates Under Increasing External Disturbances

We test extrapolation to extreme disturbances on flat terrain by independently increasing payload mass, without CoM shift, or external torso-wrench magnitude across 100 episodes per condition. Figure 3 shows that **DACT** maintains approximately 96% success at 17 kg, although **RA** and **A1** outperform it at the largest payloads. This may reflect a tradeoff between coherent torque alignment and aggressive compensation for sustained loading. Under the strongest

TABLE IV

REAL-WORLD INDOOR DISTURBANCE EXPERIMENTS COMPARING POSITION-ONLY CONTROL **DW** AND **DACT**. ROWS REPORT TRACKING, TORSO STABILITY, AND TIME-AVERAGED JOINT-WISE ℓ_1 MAGNITUDES OF COMMANDED FEEDFORWARD AND FEEDBACK TORQUES (N M). † DENOTES AN APPROACH ENCOUNTERED FAILURES DURING DATA COLLECTION.

	6L Center		8L Center		6kg Offset		8kg Offset		4kg Drop		6kg Drop		4kg Swing		6kg Swing	
Metric	DW	DC	DW	DC	DW	DC	DW	DC	DW	DC	DW	DC	DW	DC	DW	DC
$\mathbf{c}^e$	0.315	0.279	0.362	0.283	0.299	0.271	0.329†	0.297	0.211	0.183	0.237	0.196	0.237	0.234	0.231	0.233
$\mathbf{q}_j^e$	0.288	0.144	0.299	0.149	0.281	0.133	0.317†	0.148	0.203	0.094	0.211	0.107	0.196	0.087	0.191	0.094
p_z^e	0.043	0.022	0.062	0.030	0.039	0.009	0.071†	0.025	0.021	0.011	0.033	0.018	0.007	0.009	0.011	0.011
$\ \mathbf{g}\ _1$	0.076	0.062	0.081	0.063	0.095	0.059	0.137†	0.062	0.067	0.037	0.106	0.076	0.089	0.091	0.092	0.101
v_z	0.049	0.046	0.058	0.045	0.075	0.005	0.096†	0.054	0.057	0.040	0.072	0.055	0.061	0.042	0.036	0.043
$\ \boldsymbol{\omega}\ _1$	0.697	0.609	0.771	0.615	0.785	0.653	0.898†	0.758	0.548	0.487	0.661	0.612	0.893	0.716	0.935	0.981
$\ \boldsymbol{\tau}^{ff}\ _1$	—	38.14	—	44.55	—	39.05	—†	45.57	—	32.95	—	35.03	—	28.29	—	29.76
$\ \boldsymbol{\tau}^{fb}\ _1$	81.65	46.53	89.23	48.33	85.37	41.67	93.41†	44.69	60.91	30.30	63.71	32.38	58.47	32.77	59.11	34.81

impulsive disturbances, however, position-only **DW** achieves approximately 75% success, compared with 60% for **DACT** and 7% for **RA**. These results suggest that position-based feedback is particularly important for rapid recovery from abrupt disturbances. **DACT**'s robustness across both disturbance types supports retaining this stabilizing feedback structure alongside feedforward compensation, while the position-only baseline's advantage under extreme impulses highlights the importance of preserving responsive posture correction under extreme impulsive disturbances.

Dynamics Reconstruction and Consistency The internalized dynamics representation of **DACT** and **A3** are evaluated on rough terrain under payload and impulse disturbances with identical CoM-shift settings (sampled from [2.0, 1.0, 2.0]), collecting 100 episodes per rollout source and condition. We evaluate the whole-body residual dynamics as:

$$\mathbf{r}^{\text{dyn}} = \mathbf{A}\ddot{\mathbf{p}} + \mathbf{c}\dot{\rho} + \mathbf{g} - \mathbf{S}^T \boldsymbol{\tau}^{\text{app}} - \mathbf{J}_c^T \hat{\mathbf{f}}^{r,c} \quad (9)$$

Here, $\boldsymbol{\tau}^{\text{app}}$ denotes the recorded actuator torque commands from the rollout-generating policy, after combining feedforward and PD torques and applying actuator scaling and clipping. This residual measures whether predicted contact forces and recorded joint torques jointly explain the observed whole-body acceleration under the rigid-body model. Smaller residuals indicate better agreement with floating-base and joint force balance, assessing mechanical consistency beyond pointwise GRF reconstruction accuracy.

To evaluate temporal force consistency, we integrate predicted GRFs, gravity, and recorded external impulses over $T \in 50, 100$ ms. The predicted linear-momentum change is

$$\widehat{\Delta \ell} = \sum_{k=1}^K \left(\sum_i \hat{\mathbf{f}}_{i,k} + m\mathbf{g} \right) \Delta t + \mathbf{I}_{\text{ext}}, \quad K = T/\Delta t. \quad (10)$$

Angular-momentum change is calculated by integrating the corresponding moments about the fixed window-start whole-body CoM, $\mathbf{c}_0$:

$$\widehat{\Delta \mathbf{L}}^0 = \sum_{k=1}^K \left[\sum_i (\mathbf{p}_{i,k} - \mathbf{c}_0) \times \hat{\mathbf{f}}_{i,k}^{r,c} + (\mathbf{c}_k - \mathbf{c}_0) \times m\mathbf{g} \right] \Delta t + \mathbf{I}_{\text{ext}}^0 \quad (11)$$

Here, m is total robot mass including payload, $\mathbf{p}_{i,k}$ is the recorded foot-link position, and $\mathbf{c}_k$ is the whole-body CoM. External impulse terms account for the recorded momentum jumps. Comparing these predictions with observed momentum changes tests whether predicted contact forces and moments account for their accumulated effects on whole-body motion. Lower errors indicate greater consistency with linear and angular momentum balance over time, complementing instantaneous GRF reconstruction and dynamics-residual accuracy.

Table III distinguishes the *rollout source*, which generates the recorded trajectories, from the *evaluated model*, which predicts their dynamics. Within each source and disturbance condition, both models receive identical recorded histories and applied torques and are evaluated against the same simulator measurements. Consequently, comparisons within each source isolate differences in predictive accuracy on fixed transitions, while comparisons across sources reveal sensitivity to the state, actuation, and contact distributions induced by each policy.

DACT improves CoM and next-step motion-observation reconstruction on both rollout sources under both conditions. On **DACT**-generated trajectories, its GRF errors are 39.8–46.9% lower and dynamics residuals 19.0–35.3% lower than **A3**'s on identical transitions. Under impulse disturbances, the corresponding linear- and angular-momentum errors decrease by 32.3–34.9% and 25.5–43.6%, respectively, over 50–100 ms. These concurrent improvements indicate that **DACT**'s contact predictions better explain both instantaneous acceleration and accumulated momentum changes along its learned behavior. Recorded external impulses are included in both models' momentum calculations, so these comparisons assess predicted contact contributions without requiring disturbance anticipation.

Cross-source evaluation reveals a more selective advantage. On **A3**-generated impulse trajectories, **DACT**'s GRF error is 1.6% higher and its linear-momentum errors are 11.1–17.9% higher, and its 100-ms angular-momentum error is 7.8% higher. Conversely, its 100-ms angular-momentum error increases by 60.9% on **A3**-generated payload trajectories despite lower pointwise GRF error. This variation indicates

sensitivity to the motion, actuation, and contact distributions induced by each policy. Overall, the substantial reductions in force, dynamics, and momentum errors on **DACT** trajectories are consistent with the PINN objective promoting physically meaningful contact representations around the learned behavior. They also identify a limitation: contact-model accuracy does not transfer uniformly across policies' motion and actuation distributions, revealing a tight coupling between the encoder's internalized contact dynamics and the policies actions.

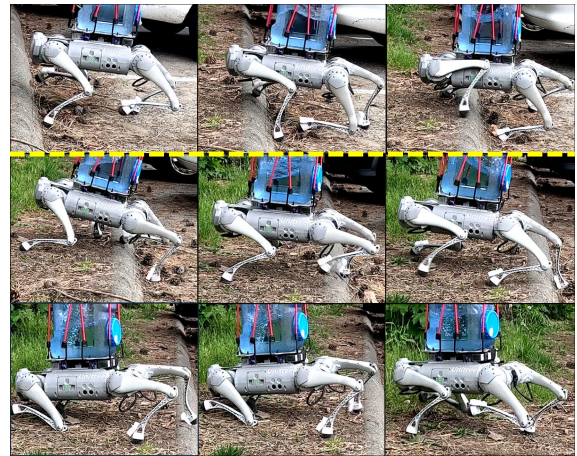
C. Real-World Experiments

Indoor Experiments:

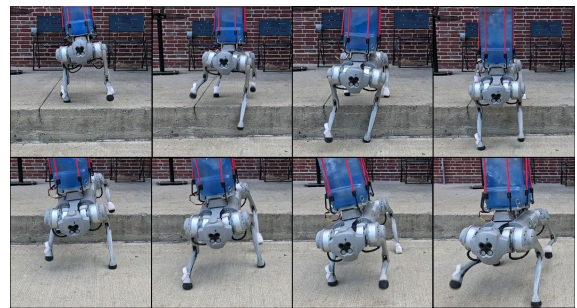
We deploy **DACT** on a Unitree Go1 using a Jetson Orin Developer Kit and compare it with the position-only baseline **DW** under sustained and transient indoor disturbances. Table IV reports tracking, torso stability, and commanded torque magnitudes for centered 6/8 L water payloads, 6/8 kg offset payloads mounted 5 cm left and 10 cm forward of the nominal CoM, and 4/6 kg masses dropped onto the robot or swung into its side using a 1 m pendulum released at 65°. Three trials are performed per condition. Centered and offset payload trials use commands within $\pm[1.0, 0.5, 1.0]$, while drop and swing trials use a 1.00 m/s forward command; swing trials additionally carry a 6 kg payload.

DACT maintains stable locomotion across the evaluated conditions, reducing command-tracking error by 1.3–21.8% in seven of eight conditions, with a small 0.9% increase under the 6 kg swing disturbance. Joint-position error decreases by 49.3–55.6% relative to **DW**, alongside improvements in nearly all reported torso-stability metrics. Under the 8 kg offset payload, the position-only baseline fails in the first two trials and again when coming to a stop in the third; its reported values use pre-failure data and therefore do not indicate successful completion. These results demonstrate the benefit of coupled torque–position control under both persistent loading and abrupt disturbances. Increasing centered and offset loads raises **DACT**'s mean feedforward torque magnitude by approximately 17%, while feedback magnitudes remain 41.1–52.2% below the baseline. These patterns suggest complementary roles: feedforward compensation addresses persistent loading, while PD feedback corrects shorter-term tracking and posture deviations, reducing reliance on sustained position errors to generate supporting forces. Indoor experiment videos are available on the project website.

Outdoor Experiments: As shown in Fig. 1, we deployed **DACT** outdoors carrying a 6.5 L liquid payload across diverse unstructured terrains. The robot successfully descended 12 cm stairs, climbed a 15° grass slope, traversed a 12° slope covered in loose pine needles and other debris, and crossed a 5° incline covered with large, loose river stones. Figure 4a highlights a challenging curb traversal in which the robot climbed a 12 cm ramped-curb despite mid-swing front-right foot contact, followed by the rear-right foot slipping off the curb and briefly catching before recovering. The robot also descended a 20 cm ledge while rejecting the sudden impact wrench induced by the sloshing liquid payload (Fig 4b). These trials demonstrate



(a) 6.5 L liquid payload over a 12 cm curb: front-foot swing contact followed by rear-foot slip.



(b) 6.5 L liquid payload down 20 cm ledge.

Fig. 4. Robust real-world locomotion with 6.5 L liquid payload.

that **DACT** transfers to real-world disturbance-rich settings, combining terrain adaptation, payload compensation, and recovery from unexpected contact; additional indoor and outdoor deployments are shown in the supplementary video.

V. CONCLUSION

We presented **DACT**, which combines coupled torque–position control, privileged dynamics reconstruction, and PINN-based whole-body regularization to align commanded torques and predicted contact forces with observed motion. Simulation and hardware experiments demonstrate robust locomotion across challenging terrains, payload and CoM variations, torso disturbances, and liquid payload transport. These results support structured action representations and physics-informed learning as a practical approach to disturbance-robust locomotion, although coupled torque control retains additional sim-to-real sensitivity associated with torque control. Future work will reduce this sensitivity through improved actuator modeling and training and enforce hard physical constraints at runtime to improve safety and reliable terrain interaction.

REFERENCES

- [1] G. Bledt, M. J. Powell, B. Katz, J. Di Carlo, P. M. Wensing, and S. Kim, “Mit cheetah 3: Design and control of a robust, dynamic quadruped robot,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.

- [2] D. Kim, J. Di Carlo, B. Katz, G. Bledt, and S. Kim, "Highly dynamic quadruped locomotion via whole-body impulse control and model predictive control," *arXiv preprint arXiv:1909.06586*, 2019.
- [3] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.
- [4] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "Rma: Rapid motor adaptation for legged robots," *Robotics: Science and Systems*, 2021.
- [5] D. Hoeller, N. Rudin, D. Sako, and M. Hutter, "Anymal parkour: Learning agile navigation for quadrupedal robots," *Science Robotics*, vol. 9, no. 88, p. eadi7566, 2024.
- [6] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.
- [7] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Conference on robot learning*. PMLR, 2022, pp. 91–100.
- [8] Z. Xiao, X. Zhang, X. Zhou, and Q. Zhang, "Pa-loco: Learning perturbation-adaptive locomotion for quadruped robots," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 9110–9115.
- [9] L. Chang, Y. Nai, H. Chen, and L. Yang, "Beyond robustness: Learning unknown dynamic load adaptation for quadruped locomotion on rough terrain," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 10 282–10 288.
- [10] S. Chen, B. Zhang, M. W. Mueller, A. Rai, and K. Sreenath, "Learning torque control for quadrupedal locomotion," in *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*. IEEE, 2023, pp. 1–8.
- [11] D. Kim, G. Berseth, M. Schwartz, and J. Park, "Torque-based deep reinforcement learning for task-and-robot agnostic learning on bipedal robots using sim-to-real transfer," *IEEE Robotics and Automation Letters*, vol. 8, no. 10, pp. 6251–6258, 2023.
- [12] P. Li, H. Li, G. Sun, J. Cheng, X. Yang, G. Bellegarda, M. Shafiee, Y. Cao, A. Ijspeert, and G. Sartoretti, "Sata: Safe and adaptive torque-based locomotion policies inspired by animal learning," *arXiv preprint arXiv:2502.12674*, 2025.
- [13] V. K. Kurva and S. Kolathaya, "Mule: Multi-terrain and unknown load adaptation for effective quadrupedal locomotion," *arXiv preprint arXiv:2505.00488*, 2025.
- [14] Y. Zhang, B. Nie, Z. Cao, Y. Fu, and Y. Gao, "Disturbance-aware adaptive compensation in hybrid force-position locomotion policy for legged robots," *arXiv preprint arXiv:2506.00472*, 2025.
- [15] S. Lyu, X. Lang, H. Zhao, H. Zhang, P. Ding, and D. Wang, "RL2ac: Reinforcement learning-based rapid online adaptive control for legged robot robust locomotion," *Robotics: Science and Systems (RSS)*, 2024.
- [16] J. Duysens, F. Clarac, and H. Cruse, "Load-regulating mechanisms in gait and posture: comparative aspects," *Physiological reviews*, vol. 80, no. 1, pp. 83–133, 2000.
- [17] T. Lam, M. Anderschitz, and V. Dietz, "Contribution of feedback and feedforward strategies to locomotor adaptations," *Journal of neurophysiology*, vol. 95, no. 2, pp. 766–773, 2006.
- [18] S. Anwer, H. Li, M. F. Antwi-Afari, W. Umer, I. Mehmood, and A. Y. L. Wong, "Effects of load carrying techniques on gait parameters, dynamic balance, and physiological parameters during a manual material handling task," *Engineering, Construction and Architectural Management*, vol. 29, no. 9, pp. 3415–3438, 2022.
- [19] M. V. Minniti, R. Grandia, F. Farshidian, and M. Hutter, "Adaptive clf-mpc with application to quadrupedal robots," *IEEE Robotics and Automation Letters*, vol. 7, no. 1, pp. 565–572, 2021.
- [20] M. Somboldestan, Y. Chen, and Q. Nguyen, "Adaptive force-based control for legged robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.
- [21] L. Amanzadeh, T. Chunawala, R. T. Fawcett, A. Leonessa, and K. A. Hamed, "Predictive control with indirect adaptive laws for payload transportation by quadrupedal robots," *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 10 359–10 366, 2024.
- [22] R. Wang, S. Lyu, and D. Wang, "Dynamic adaptive legged locomotion policy via decoupling reaction force control and gait control," *arXiv preprint arXiv:2509.13737*, 2025.
- [23] G. Tournois, M. Focchi, A. Del Prete, R. Orsolino, D. G. Caldwell, and C. Semini, "Online payload identification for quadruped robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017.
- [24] Z. Zhu, G. Zhang, Z. Sun, T. Chen, X. Rong, A. Xie, and Y. Li, "Proprioceptive-based whole-body disturbance rejection control for dynamic motions in legged robots," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7703–7710, 2023.
- [25] M. Somboldestan and Q. Nguyen, "Adaptive-force-based control of dynamic legged locomotion over uneven terrain," *IEEE Transactions on Robotics*, vol. 40, pp. 2462–2477, 2024.
- [26] A. Trivedi, S. Prajapati, M. Zolotas, M. Everett, and T. Padi, "Chance-constrained convex mpc for robust quadruped locomotion under parametric and additive uncertainties," *IEEE Robotics and Automation Letters*, 2025.
- [27] I. M. A. Nahrendra, B. Yu, and H. Myung, "Dreamwaq: Learning robust quadrupedal locomotion with implicit terrain imagination via deep reinforcement learning," in *2023 IEEE International Conference on Robotics and Automation, ICRA 2023*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 5078–5084.
- [28] Y. Chen and Q. Nguyen, "Learning agile locomotion and adaptive behaviors via rl-augmented mpc," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [29] X. Zeng, H. Zhang, L. Yue, Z. Song, L. Zhang, and Y.-H. Liu, "Adaptive model predictive control with data-driven error model for quadrupedal locomotion," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [30] S. Lyu, H. Zhao, and D. Wang, "A composite control strategy for quadruped robot by integrating reinforcement learning and model-based control," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 751–758.
- [31] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational physics*, vol. 378, pp. 686–707, 2019.
- [32] J.-Q. Wang, L. Song, J. Shen, B. Yong, X. Han, Y. Jiang, M. Raoofi, and Q. Zhou, "Physics-informed continuous-time reinforcement learning with data-driven approach for robotic arm manipulation," *Journal of Industrial Information Integration*, p. 101008, 2025.
- [33] Z. Mao, A. D. Jagtap, and G. E. Karniadakis, "Physics-informed neural networks for high-speed flows," *Computer Methods in Applied Mechanics and Engineering*, vol. 360, p. 112789, 2020.
- [34] D. Jalili, S. Jang, M. Jadidi, G. Giustini, A. Keshmiri, and Y. Mahmoudi, "Physics-informed neural networks for heat transfer prediction in two-phase flows," *International Journal of Heat and Mass Transfer*, vol. 221, p. 125089, 2024.
- [35] S. Sanyal and K. Roy, "Ramp-net: A robust adaptive mpc for quadrotors via physics-informed neural network," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [36] W. Gu, S. Primatesta, and A. Rizzo, "Physics-informed neural network for quadrotor dynamical modeling," *Robotics and Autonomous Systems*, vol. 171, p. 104569, 2024.
- [37] M. Lahariya, C. Innes, C. Develder, and S. Ramamoorthy, "Learning physics-informed simulation models for soft robotic manipulation: A case study with dielectric elastomer actuators," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [38] X. Yang, Y. Du, L. Li, Z. Zhou, and X. Zhang, "Physics-informed neural network for model prediction and dynamics parameter identification of collaborative robot joints," *IEEE Robotics and Automation Letters*, vol. 8, no. 12, pp. 8462–8469, 2023.
- [39] I. Sorrentino, G. Romualdi, L. Moretti, S. Traversaro, and D. Pucci, "Physics-informed neural networks with unscented kalman filter for sensorless joint torque estimation in humanoid robots," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5705–5712, 2025.
- [40] R. Ni and A. H. Qureshi, "Physics-informed neural motion planning on constraint manifolds," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [41] J. Carpentier and N. Mansard, "Analytical derivatives of rigid body dynamics algorithms," in *Robotics: Science and systems (RSS 2018)*, 2018.
- [42] G. Authors, "Genesis: A generative and universal physics engine for robotics and beyond," December 2024. [Online]. Available: <https://github.com/Genesis-Embodied-AI/Genesis>